

# Static Gesture Classification — Run Report

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## Data paths

|                      |                                                                  |
|----------------------|------------------------------------------------------------------|
| Training data path 1 | /home/jestin/ThesisRepo/ML/NewTrainingData/Static · 365 trials   |
| Total training paths | 1 · 365 trials total                                             |
| External test root   | /home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static           |
| Report output dir    | /home/jestin/ThesisRepo/ML/saved_reports/AugmentationExperiments |

## Sensor selection

|                |                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------|
| Hands          | left                                                                                                |
| Segments       | palm_mid, thumb, index, middle, ring, pinky, wrist                                                  |
| Modalities     | YPR, Accel, Flex                                                                                    |
| Sensor columns | 88 channels                                                                                         |
| Classes        | 6: Double_Closed_Fist, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman |

## Preprocessing, features & split

|                             |                                             |
|-----------------------------|---------------------------------------------|
| Resample steps              | 90                                          |
| Butterworth                 | off · cutoff 10.0 Hz · order 4 · fs 30.0 Hz |
| Normalisation               | minmax                                      |
| Feature mode                | stats+fft → feature dim 4752                |
| Test size · seed · CV folds | 0.2 · random_state=42 · 10-fold stratified  |
| Sample counts               | train (post-aug)=584 · test=73              |

## Augmentation (training only)

|                        |                        |
|------------------------|------------------------|
| Enabled                | yes                    |
| Copies per train trial | 1                      |
| Jitter                 | off · sigma=0.01       |
| Amplitude scale        | on · range=(0.8, 1.2)  |
| Time warp              | off · range=(0.9, 1.1) |

## Classifiers — cross-validation & hold-out test

| Algorithm                       | CV mean (10-fold) | CV std | Hold-out test acc |
|---------------------------------|-------------------|--------|-------------------|
| SVM (RBF)                       | 0.9983            | 0.0052 | 1.0000            |
| SVM (Linear)                    | 1.0000            | 0.0000 | 0.9863            |
| Logistic Regression             | 1.0000            | 0.0000 | 0.9863            |
| Voting Soft (SVM+RF+KNN+LogReg) | 1.0000            | 0.0000 | 0.9863            |
| Stacking (SVM+RF+KNN+LogReg)    | 1.0000            | 0.0000 | 0.9863            |
| Random Forest                   | 1.0000            | 0.0000 | 0.9726            |
| KNN                             | 0.9709            | 0.0219 | 0.9178            |
| LDA                             | 0.9761            | 0.0218 | 0.9178            |

## Per-class hold-out F1

| Class              | SVM (RBF) | SVM (Linear) | Logistic Regression | Voting (soft) | Stacking | Random Forest | KNN   | LDA   | Support |
|--------------------|-----------|--------------|---------------------|---------------|----------|---------------|-------|-------|---------|
| Double_Closed_Fist | 1.000     | 1.000        | 1.000               | 1.000         | 1.000    | 0.963         | 0.786 | 0.846 | 13      |
| Double_Okay        | 1.000     | 1.000        | 1.000               | 1.000         | 1.000    | 0.957         | 0.957 | 0.917 | 12      |
| Double_Open_Palm   | 1.000     | 1.000        | 1.000               | 1.000         | 1.000    | 1.000         | 0.960 | 0.957 | 12      |
| Double_Phone       | 1.000     | 1.000        | 1.000               | 1.000         | 1.000    | 1.000         | 0.818 | 0.889 | 12      |
| Double_Pistol      | 1.000     | 0.960        | 0.960               | 0.960         | 0.960    | 0.960         | 1.000 | 0.957 | 12      |
| Double_Spiderman   | 1.000     | 0.957        | 0.957               | 0.957         | 0.957    | 0.957         | 1.000 | 0.957 | 12      |
| macro avg          | 1.000     | 0.986        | 0.986               | 0.986         | 0.986    | 0.973         | 0.920 | 0.920 | 73      |

## External test (NewTestData)

| Algorithm                       | External test acc |
|---------------------------------|-------------------|
| Random Forest                   | 0.8710            |
| Stacking (SVM+RF+KNN+LogReg)    | 0.8065            |
| Voting Soft (SVM+RF+KNN+LogReg) | 0.7742            |
| SVM (Linear)                    | 0.7097            |
| Logistic Regression             | 0.7097            |
| SVM (RBF)                       | 0.6774            |
| LDA                             | 0.6452            |
| KNN                             | 0.5484            |

## Per-class external accuracy

| Class              | Random Forest | Stacking | Voting (soft) | SVM (Linear) | Logistic Regression | SVM (RBF) | LDA   | KNN   | Support |
|--------------------|---------------|----------|---------------|--------------|---------------------|-----------|-------|-------|---------|
| Double_Closed_Fist | 1.000         | 1.000    | 0.833         | 0.333        | 0.667               | 0.500     | 0.500 | 0.167 | 6       |
| Double_Okay        | 0.200         | 0.000    | 0.000         | 0.000        | 0.000               | 0.000     | 0.000 | 0.000 | 5       |
| Double_Open_Palm   | 1.000         | 1.000    | 1.000         | 1.000        | 1.000               | 1.000     | 0.600 | 1.000 | 5       |
| Double_Phone       | 1.000         | 0.800    | 0.800         | 1.000        | 0.600               | 0.600     | 1.000 | 0.200 | 5       |
| Double_Pistol      | 1.000         | 1.000    | 1.000         | 1.000        | 1.000               | 1.000     | 1.000 | 1.000 | 5       |
| Double_Spiderman   | 1.000         | 1.000    | 1.000         | 1.000        | 1.000               | 1.000     | 0.800 | 1.000 | 5       |